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Nitte (Deemed to be University)

(Established under Sec (3) of UGC ACT,1956)

First Semester MCA (CBCS) Degree Examinations, Dec. 2025 - Jan. 2026**25MCA106 – ADVANCED OPERATING SYSTEMS WITH UNIX**

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain any 2 types of Advanced Operating Systems.	10	L2	1	1
b)	Discuss any five key features of advanced operating system.	10	L2	1	1
2. a)	Describe the mechanisms of inter-process communication (IPC) and also explain 2 models used in IPC.	10	L2	1	1
b)	Explain types of semaphores with suitable examples.	10	L2	1	1
Unit – II					
3. a)	Discuss the Kernel Components used in UNIX Operating System.	10	L2	2	2
b)	Analyze how system calls and the shell act as an interface between user and kernel.	10	L3	2	2
4. a)	Explain the structure of the UNIX file system. How are inodes and system calls used in file management.	10	L2	2	2
b)	Analyze how memory management and virtual memory mechanisms works in UNIX.	10	L3	2	2
Unit – III					
5. a)	Outline the Core Components of a Distributed System.	10	L2	3	2
b)	Create a comparison table highlighting key differences between Distributed and Centralized Systems.	10	L3	3	2
6. a)	Illustrate the Architectural model of distributed OS along with its three broad styles	10	L2	3	2
b)	Analyze the need for scheduling in Distributed Operating Systems and also explain two Scheduling Strategies.	10	L3	3	2
Unit – IV					
7. a)	Discuss the security mechanisms in UNIX including access control and firewalls.	10	L2	4	2
b)	Analyze how intrusion detection and prevention systems works in advanced operating environments.	10	L3	4	2
8. a)	Summarize various performance optimization techniques in UNIX. How does process scheduling affect performance in multi-core systems?	10	L2	4	2
b)	Analyze resource management strategies that improve performance in advanced operating systems.	10	L3	4	2
Unit – V					
9. a)	Analyze how Docker and Kubernetes enhance scalability, resource optimization, and deployment efficiency in cloud-based environments.	10	L2	5	2
b)	Explain the characteristics and applications of Real-Time Operating Systems (RTOS) in embedded environments.	10	L3	5	2

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| 10. | a) | Create a case study on virtualization in cloud environments focusing on scalability and fault tolerance. | 10 | L2 | 5 | 1 |
| | b) | Evaluate the challenges and opportunities of AI-driven resource management in next-generation operating systems. | 10 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

(Max. Marks: 100)

Unit - I						
Sl. No.	CO	BT	Level	Marks	PO	PO
1.	1	L2	1	1	1	1
2.	1	L2	1	1	1	1
Unit - II						
3.	2	L2	2	2	2	2
4.	2	L3	2	2	2	2
Unit - III						
5.	3	L2	3	3	3	3
6.	3	L3	3	3	3	3
Unit - IV						
7.	4	L2	4	4	4	4
8.	4	L3	4	4	4	4
Unit - V						
9.	5	L2	5	5	5	5
10.	5	L3	5	5	5	5